

Vision Transformers for Breast Cancer Prediction in Structural Mammography and Functional Infrared Thermography

Mi tesis en 180 segundos — ICI2ST 2026

Lenin G. Falconí

Escuela Politécnica Nacional

Advisors: PhD. María Pérez (EPN) & PhD. Aura Conci (UFF)

2026-09-25

Problem Context

- Breast cancer is a **leading cause of cancer death** in women, in Ecuador and worldwide.
- Global incidence is expected to exceed 6 million new cases by 2050.
- **Early detection saves lives**, but reading images is slow and prone to miss subtle findings.

Female Cancer Mortality in Ecuador

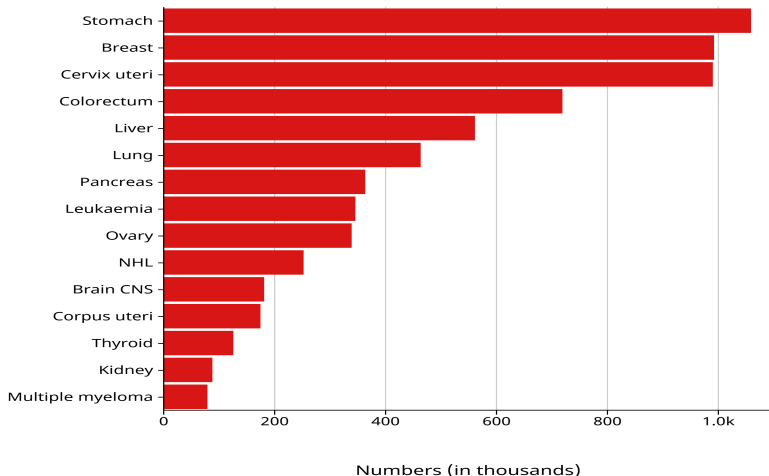


Figure: Female cancer mortality in Ecuador (Global Cancer Observatory IARC,2024)

- Vision transformers are **little studied for infrared thermography**; public datasets **rarely pair** mammography and thermography for the same patient; and **no standardized benchmarking** exists, making results hard to compare.
- **Gap 1 — scarce pectoral-muscle annotations** in mammography: zero-shot MedSAM study [1].
- **Gap 2 — no per-view thermography benchmark**: view-conditioned SegFormer study [2].
- **Gap 3 — temporal dynamics underexplored in dynamic infrared**: TimeSformer study [3].
- *Proposal*: one transformer-based model per gap, trained and evaluated under a common patient-level cross-validation protocol.

Research Goal and Proposal

Proposal (Hypothesis)

*Under controlled training and evaluation protocols, transformer-based models **match or outperform** their convolutional baselines for breast-cancer prediction in structural mammography and functional thermography.*

Goal

To develop and evaluate transformer-based models for the semantic segmentation of breast lesions in mammography, and for region-of-interest segmentation and malignancy classification of thermal patterns in infrared thermography, using a quantitative, quasi-experimental Design Science Research protocol with patient-level cross-validation.

Video Transformers for Dynamic Infrared Thermography (Published, 2024)

- Fine-tuned **TimeSformer** on the 20-frame dynamic sequences of **DMR-UFF** (149 patients) [3].
- Learns **temporal thermal patterns** under a controlled protocol of thermal stress applied to patient [4], [5]:

Study	Model	Protocol	Unit	AUC
Garia et al. (2023)	ViT	Static (SIT)	Image (950)	0.9570
Baffa et al. (2023)	3D-CNN	Dynamic (DIT)	Patient (149)	0.9351
Ours	TimeSformer (ViViT)	Dynamic (DIT)	Patient (149)	0.9444

- *Impact:* **best AUC reported on the dynamic protocol** — +0.9 points over the 3D-CNN baseline on a held-out 15-patient test set, using **fewer frames** (8 of 20) at **higher resolution** (224×224 vs 32×32) [3]. Published in Springer LNCS (AIIMA 2025).

Video Transformers for Dynamic Infrared Thermography (Published, 2024)

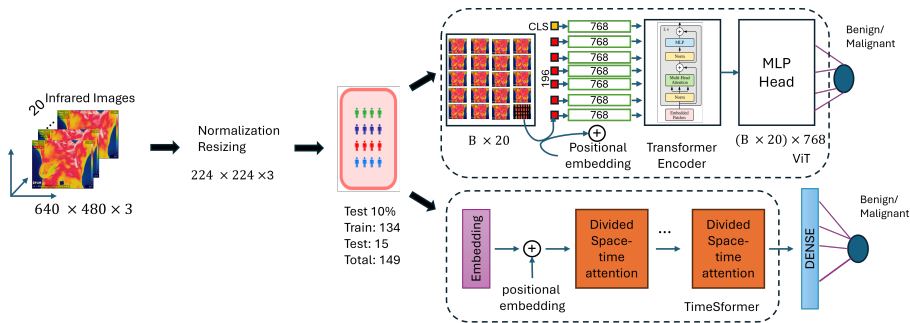


Figure: Vision Transformer pipeline used in this research [3].

Zero-Shot Learning for Pectoral Muscle Segmentation with MedSAM (Accepted, 2025)

- Ground-truth pectoral-muscle masks are **scarce**. We tested a **zero-shot foundation model**: **MedSAM** prompted by a human expert [1].
- On 168 MIAS cases: mean DSC = 0.865, mean IoU = 0.783, median DSC = 0.909 [1].
- Accepted for publication (proceedings pending indexing).

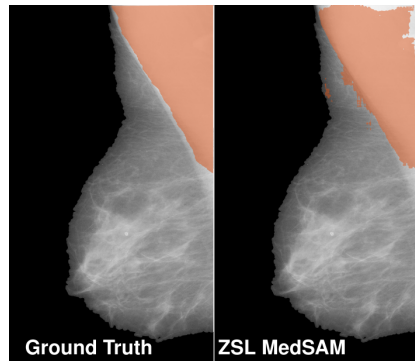


Figure: Pectoral muscle overlay [1].

ROI Segmentation in Thermography (Under Review, 2026)

- One lightweight SegFormer-B0 with a view-conditioning module segments all five canonical views (frontal, oblique 45° , lateral 90° , each side) [2].
- Compared with published DMR–UFF baselines [6]–[8], it improves on both frontal and lateral views under patient-level 5-fold CV:

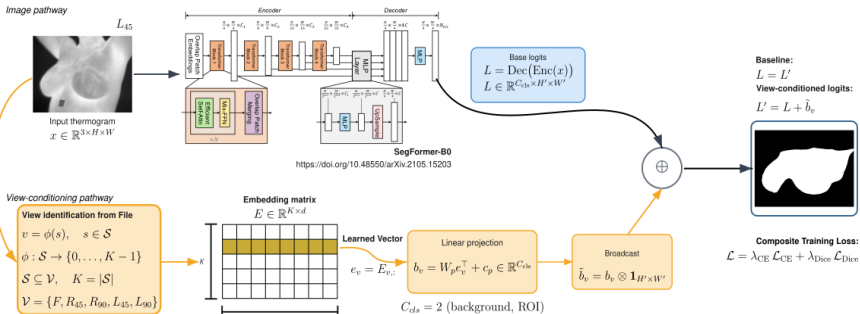
Study	Model	View	IoU
Baffa et al. (2021)	U-Net	Frontal	94.38%
Carvalho et al. (2023)	U-Net	Lateral	82.30%
Jaincy et al. (2025)	DSU-Net-SCA	Frontal	88.50%
Ours	SegFormer-B0	Frontal	96.66%
Ours	SegFormer-B0	Lateral	85.31%
Ours	SegFormer-B0 + VC ¹	All 5	92.61%

- *Impact*: +2.3 IoU points over the best frontal baseline and +3.0 over the lateral baseline — the first per-view benchmark on DMR–UFF, with a single model for all five views [2].

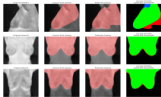
¹VC: View Conditioned

ROI Segmentation in Thermography (Under Review, 2026)

View-Conditioned SegFormer-B0: Post-Decoder Logit-Bias Architecture



Qualitative Results:



Quantitative Results:

- Overall: 92.61%
- Frontal: 96.66%
- Lateral: 88.98%

Figure: Proposed post-decoder logit-bias view-conditioning [2].

Conclusions and Future Work I

- **Thermography — classification**: video transformers are comparable to the 3D-CNN baseline ($\text{AUC} = 0.94$) on dynamic infrared thermography; the role of temporal dynamics still needs further study [3].
- **Mammography — segmentation**: zero-shot MedSAM segments the pectoral muscle (median DSC = 0.909) without task-specific training, aiding annotation but not replacing expert ground truth [1].
- **Thermography — segmentation**: one lightweight SegFormer-B0 with view-conditioning segments all five canonical views ($\text{IoU} = 0.9261$ under patient-level 5-fold CV) — the first per-view benchmark on DMR-UFF [2].

Conclusions and Future Work II

- **Future work (Second DSR iteration):** (i) chain the thermography SegFormer ROI with the video transformer classifier, (ii) compare zero-shot MedSAM with a fine-tuned variant in mammography, and (iii) perform a generalization study of current models on a second unseen dataset for infrared thermography.

Thank You

- PhD Program in Informatics — Escuela Politécnica Nacional
- Advisors: PhD. María Gabriela Pérez Hernández (EPN) and PhD. Aura Conci (UFF).
- Contact:
 - email: lenin.falconi@epn.edu.ec
 - ORCID: 0000-0003-4402-6643
 - linkedin: linkedin.com/in/lenin-g-falconi
 - github: github.com/LeninGF

- [1] L. G. Falconí and M. Pérez. “Towards a segmentation of the pectoral muscle and masses present in mammograms by combining modern and traditional techniques.” *Accepted; proceedings pending indexing*. 2026.
- [2] L. G. Falconí, M. Pérez, and A. Conci. “View-Conditioned SegFormer-B0 for Multi-View Breast Thermography ROI Segmentation.” *Manuscript under review, IEEE Journal of Biomedical and Health Informatics*. 2026.
- [3] L. G. Falconí, M. Pérez, M. E. Benalcázar, et al. “Video Transformers for Dynamic Breast Infrared Imaging Classification.” In: *Artificial Intelligence over Infrared Images for Medical Applications (AIIIMA 2025)*. Lecture Notes in Computer Science. Springer, 2025, pp. 1–19. DOI: [10.1007/978-3-032-10990-3_1](https://doi.org/10.1007/978-3-032-10990-3_1).

References II

- [4] L. S. Garia and M. Hariharan. “Vision Transformers for Breast Cancer Classification from Thermal Images.” In: *Lecture Notes in Electrical Engineering* 1009 (2023), pp. 177–185. DOI: [10.1007/978-981-99-0236-1_13](https://doi.org/10.1007/978-981-99-0236-1_13).
- [5] M. de Freitas Oliveira Baffa, L. G. Lattari, and A. Conci. “3D Convolutional Neural Networks for Dynamic Breast Infrared Imaging Classification.” In: *Artificial Intelligence over Infrared Images for Medical Applications (AIIMA 2023)*. Lecture Notes in Computer Science. Springer, 2023, pp. 57–66. DOI: [10.1007/978-3-031-44511-8_4](https://doi.org/10.1007/978-3-031-44511-8_4).
- [6] M. d. F. O. Baffa, A. M. Coelho, and A. Conci. “Segmentação de Imagens Infravermelhas Para Detecção do Câncer de Mama Utilizando U-Net CNN.” In: *Simpósio Brasileiro de Computação Aplicada à Saúde (SBCAS)*. SBC, 2021, pp. 119–128. DOI: [10.5753/sbcas.2021.16058](https://doi.org/10.5753/sbcas.2021.16058).

- [7] E. C. de Carvalho, A. M. Coelho, A. Conci, et al. “U-Net convolutional neural networks for breast IR imaging segmentation on frontal and lateral view.” In: *Computer Methods in Biomechanics and Biomedical Engineering: Imaging and Visualization* 11.3 (2023), pp. 311–316. DOI: [10.1080/21681163.2022.2040053](https://doi.org/10.1080/21681163.2022.2040053).
- [8] D. E. Martina Jaincy and V. Pattabiraman. “Deep Swin U-Net with Spatial and Channel Attention for Breast Tumor Segmentation in Infrared Images.” In: *2025 International Conference on Sensors and Related Networks (SENNET 2025)*. IEEE, 2025. DOI: [10.1109/SENNET64220.2025.11135940](https://doi.org/10.1109/SENNET64220.2025.11135940).